D 92282 (Pages : 3) Name.....

Reg. No.....

THIRD SEMESTER B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

(CUCBCSS—UG)

Complementary Course

MAT 3C 03—MATHEMATICS

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type Questions)

Answer all **twelve** *questions.*

1. Write the general form of Bernoulli's differential equation.
2. Find the solution of the differential equation $y' = \frac{-y}{x}$.
3. What is the order of the differential equation $y'' - (y')^3 + 4 = 0$?
4. State Cayley Hamilton theorem.
5. What is the rank of a $(n \times n)$ non-singular matrix ?
6. Write the normal form of the matrix : $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
7. Write the parametric equation of the curve $\frac{x^2}{4} + \frac{y^2}{3} = 1$.
8. Define Irrotational vector.
9. Find curl v, where $v = [2y, 5x, 0]$.
10. Find the tangent to the curve $r(t) = ti + t^3 j$ at $(1, 1, 0)$.
11. Define scalar potential of a vector.
12. State Gauss's divergence theorem.

(12 × 1 = 12 marks)

Part B (Short Answer Type Questions)

Answer any **nine** *questions.*

13. Find the orthogonal trajectories of the family of curves $y = ce^{-x}$.
14. Write the condition for the differential equation $Mdx + Ndy = 0$ become exact. What is the form of its solution ?

Turn over

15. Find the integrating factor of the linear differential equation $y' - y = e^{2x}$.

16. Find characteristic roots of the matrix : $\begin{bmatrix} -1 & 2 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.

17. Write the elementary transformations in a matrix.

18. Find the comment of vector $a = [4, 2, 0]$ in the direction of $b = [1, -1, 2]$.

19. Find the directional derivative of $f = xyz$ at the point P $(-1, 1, 3)$ in the direction of $i - 2j + 2k$.

20. Find the unit normal to the level surface $z^2 = 4(x^2 + y^2)$ at the point P $(1, 0, 2)$.

21. Find div v, where $v = xyzi + 3zxj + zk$.

22. Define Jacobian.

23. Find value of λ if $a = [4, 2, \lambda]$ and $b = [2, -3, 1]$ are orthogonal.

24. Write the formula for finding the area of a plane region as a line integral over the boundary.

(9 × 2 = 18 marks)

Part C (Short Essay Type Questions)

Answer any **six** *questions.*

25. Solve the intial value problem $y' + y \tan x = \sin 2x, y(0) = 1$.

26. Solve $xy' = y + 3x^4 \cos^2\left(\frac{y}{x}\right)$.

27. Find the eigenvalues and eigenvector corresponding to any one eigenvalue of the matrix : $A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 4 & 0 \\ -2 & 0 & 5 \end{bmatrix}$.

28. Use Cayley Hamilton theorem to find A^{-1} and A^4, where $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$.

29. Find the tagential and normal componets of acceleration of an object moving along the curve $r(t) = e^t i + e^{-1} j$.

30. Find tangent to the ellipse $\frac{1}{4}\left(x^2 + y^2\right)$ at the point $P\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

31. Find the area of the cardioid $r = a(1 - \cos\theta)$, $0 \le \theta \le 2\pi$.

32. Evaluate the double integral $\iint_R y^2\, dxdy$ where R is the region bounded by the unit circle in the first quadrant.

33. Verify Green's theorem in the plane for the vector $F = \left(y^2 - 7y\right)i + (2xy + 2x)\, j$ and the region bounded by the $x^2 + y^2 = 1$.

(6 × 5 = 30 marks)

Part D (Essay Type Questions)

Answer any **two** *questions.*

34. Test for consistency and solve the following system of equation.

(a)
$$\begin{aligned} x + y + z + 3 &= 0 \\ 3x + y - 2z + 2 &= 0 \\ 2x + 4y + 7z - 13 &= 0. \end{aligned}$$

(b)
$$\begin{aligned} 3x + 26y + 2z &= 9 \\ 5x + 3y + 7z &= 4 \\ 7x + 2y + 10z &= 5. \end{aligned}$$

35. (a) Solve the differential equation :

$$2\sin\left(y^2\right)dx + xy\cos\left(y^2\right)dy = 0,\ y(2) = \sqrt{\frac{\pi}{2}}.$$

(b) Prove that Curl(gradf) = 0.

36. Verify Stokes's theorem for $F = [y, z, x]$ over the surface of the paraboloid $z = 1 - \left(x^2 + y^2\right)$, $z \ge 0$.

(2 × 10 = 20 marks)